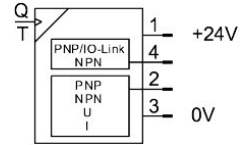


Flow sensor SFAB-

Part number: 563795

FESTO



[PDF](#) General operating condition

Data sheet

Overall data sheet – Individual values depend upon your configuration.

Feature	Value
Symbol	00992242 00992243 00995566
Certification	RCM compliance mark c UL us - Listed (OL)
CE marking (see declaration of conformity)	As per EU EMC directive as per EU explosion protection directive (ATEX) As per EU RoHS directive
UKCA marking (see declaration of conformity)	To UK instructions for EMC To UK RoHS instructions
Explosion prevention and protection	Zone 2 (ATEX) Zone 22 (ATEX)
ATEX category gas	II 3G
Type of ignition protection for gas	Ex nA IIC T5 X Gc
ATEX category for dust	II 3D
Type of (ignition) protection for dust	Ex tc IIIB T80°C X Dc IP54
Explosive ambient temperature	0°C ≤ Ta ≤ +50°C
Certificate issuing authority	UL E322346
Note on materials	RoHS-compliant
Measured variable	Mass flow rate Temperature Volume Volumetric flow rate
Flow direction	Unidirectional P1 → P2
Measuring principle	Thermal
Method of measurement	Heat loss
Flow measuring range start value	0.1 l/min ... 10 l/min
Flow measuring range end value	3 l/min ... 1000 l/min
Temperature measuring range start value	0 °C
Temperature measuring range end value	50 °C
Operating pressure	0 MPa ... 1 MPa
Operating pressure	0 bar ... 10 bar
Operating pressure	0 psi ... 145 psi

Feature	Value
Operating medium	Argon Compressed air as per ISO 8573-1:2010 [6:4:4] Compressed air as per ISO 8573-1:2010 [7:4:4] Carbon dioxide Nitrogen
Temperature of medium	0 °C ... 50 °C
Ambient temperature	0 °C ... 50 °C
Nominal temperature	23 °C
Accuracy of flow rate	± (3% o.m.v. + 0.3% FS)
Accuracy of temperature in ± °C	5 °C
Zero point repetition accuracy in ± %FS	0.2 %FS
Repetition accuracy margin in ± %FS	0.8 %FS
Temperature co-efficient margin in ± %FS/K	typ. 0. 1%FS/K
Pressure influence of margin in ±%FS/bar	0.5 %FS/b.
Switching output	2 x PNP or 2 x NPN switchable
Switching function	Window comparator Threshold value comparator
Switching element function	N/C contact/N/O contact switchable
On time	10 ms
Switch-off time	10 ms
Max. output current	100 mA
Analog output	0 - 10 V 4 - 20 mA 1 - 5 V
Flow characteristic curve, start value	0 l/min
Flow characteristic curve, end value	10 l/min ... 1000 l/min
Temperature characteristic curve start value	0 °C
Temperature characteristic curve end value	100 °C
Output characteristic curve initial value	0 V
End value output characteristic curve	10 V
Output characteristic curve initial value	4 mA
End value output characteristic curve	20 mA
Max. load resistance of current output	500 Ohm
Min. load resistance of voltage output	10 kOhm
Short-circuit protection	yes
Overload protection	Available
Protocol	IO-Link®
IO-Link, revision ID	V1.1
IO-Link, device profile	Function Extended identification Function Measurement data, standard resolution Function Multiple switching signal Firmware update Function Locator Function Product URI Function Teach single value Identification and diagnosis Smart Sensor - SSP 4.1.2
IO-Link, transmission rate	COM3
IO-Link®, SIO mode support	Yes
IO-Link, port type	Class A
IO-Link, process data length output	0 bit
IO-Link, process data length input	64 bit
IO-Link®, process data content IN	Flow rate measured value 16-bit MDC Flow rate monitoring 2-bit SSC Temperature measured value 16-bit MDC Temperature monitoring 2-bit SSC Volume / mass pulse 1 bit SSC
IO-Link®, service data contents IN	Volume / mass measured value 32-bit

Feature	Value
IO-Link®, minimum cycle time	1.2 ms
IO-Link®, data memory required	500 byte
DC operating voltage range	15 V ... 30 V
Idle current	90 mA ... 120 mA
Reverse polarity protection	for all electrical connections
Electrical connection 1, connection type	Plug
Electrical connection 1, connection technology	M12x1 A-coded as per EN 61076-2-101
Electrical connection 1, number of pins/wires	5
Electrical connection 1, type of mounting	Screw-type lock not rotatable
Electrical connection 1, type of mounting	Compatible with screw lock rotatable
Electrical connection for input 1, connection pattern	00995383
Type of mounting	With through-hole With H-rail With wall/surface bracket
Mounting position	Any
Pneumatic connection	For pneumatic tubing outside diameter 6 mm For pneumatic tubing outside diameter 8 mm For pneumatic tubing outside diameter 10 mm For tubing O.D. 12 mm for tubing O.D. 1/4" for tubing O.D. 5/16" for tubing O.D. 3/8"
Product weight	160 g
Housing material	PA-reinforced
Display type	Illuminated LCD, multi-color
Displayable unit(s)	g g/min l l/h l/min m3 m3/h scf scfm
Setting options	IO-Link® Teach-in Via display and pushbuttons
Protection against tampering	IO-Link® PIN code
Degree of protection	IP65
Pressure drop	<100 mbar
Protection class	III
Corrosion resistance class (CRC)	2 - Moderate corrosion stress
LABS (PWIS) conformity	VDMA24364-B1/B2-L